Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A laser beam of power $27\ \mathrm{mW}$ has a cross-sectional area of $10\ \mathrm{mm}^2$. The magnitude of the maximum electric field in this electromagnetic wave is given by [Take $\varepsilon_0 \approx 9 \times 10^{-12}\ \mathrm{C}^2\ \mathrm{N}^{-1}\ \mathrm{m}^{-2}$ and $c = 3 \times 10^8\ \mathrm{m\ s}^{-1}$]

(A) $1\ \mathrm{kV/m}$

(B) $1.4\ \mathrm{kV/m}$

(C) $0.7\ \mathrm{kV/m}$

(D) $2\ \mathrm{kV/m}$

Q.2

Which of the following statements is/are correct for an x-ray tube ?

(A) On increasing potential difference between filament and target, current of x-rays decreases

(B) On increasing potential difference between filament and target, frequency of x-rays increases

(C) On increasing filament current, cut-off wavelength increases

(D) None of the above.

Q.3

Electric and magnetic field are directed as $\mathrm{E_0\hat{i}}$ and $\mathrm{B_0\hat{k}}$, a particle of mass m and charge $+\mathrm{q}$ is released from position $(0, 2, 0)$ from rest. The velocity of that particle at $(x, 5, 0)$ is $(5\hat{i} + 12\hat{j})$ the value of x can be

(A) $\frac{169\ \mathrm{m}}{2q\mathrm{E_0}}$

(B) $\frac{25m}{2qE_0}$

(C) $\frac{25m}{12qE_0}$

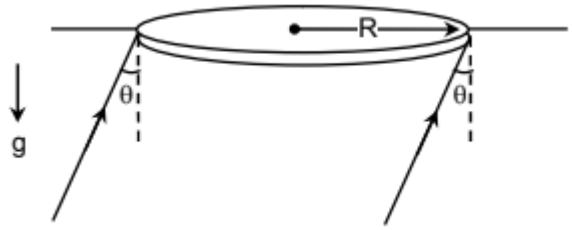
(D) $\frac{144\ \mathrm{m}}{12q\mathrm{E_0}}$

Q.4

A uniform disc of mass m and Radius R is to be kept in equilibrium such that its one surface is touched at a horizontal wall, by irradiating a parallel beam of electromagnetic radiation of intensity I, at an angle θ with the vertical. Whole arrangement is shown in figure. Disc is perfectly absorbing.

If the minimum coefficient of friction required between disc and horizontal wall for this propose is $\mu = \frac{\alpha I\pi R^2}{\beta I\pi R^2 - \gamma \mathrm{mgC}}$. Find minimum value of $\alpha + \beta + \gamma$. (use $\theta = \frac{\pi}{4}$).

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(A) 5

(B) 10

(C) 8

(D) 4

Q.5

The magnetic field of a plane electromagnetic wave is given by $\vec{B} = B_0\hat{i}[\cos(kz - \omega t)] + B_1\hat{j}\cos(kz + \omega t)$, where $B_0 = 3 \times 10^{-5}$ T and $B_1 = 2 \times 10^{-6}$ T. The RMS value of the force experienced by a stationary charge $Q = 10^{-4}$ C at $z = 0$ is closest to

(A) 0.1 N

(B) 0.9 N

(C) 3×10^{-2} N

(D) 0.6 N

Q.6

A 50 W m^{-2} energy density of sunlight is incident normally on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1 m^2 surface area will be close to $\left(c = 3 \times 10^8 \text{ m s}^{-1}\right)$

(A) 15×10^{-8} N

(B) 20×10^{-8} N

(C) 10×10^{-8} N

(D) 35×10^{-8} N

Q.7

A charged dust particle of radius 5×10^{-7} m is located in a horizontal electric field having an intensity of 6.28×10^5 V m^{-1}. The surrounding medium in air with coefficient of viscosity $\eta = 1.6 \times 10^{-15}$N s m^{-2}. If this particle moves with a uniform horizontal speed of 0.01 m s^{-1}, the number of electrons on it will be

(A) 20

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(B) 15

(C) 25

(D) 30

Q.8

In X-ray tube, when the accelerating voltage V is halved, what will be the difference between the wavelength of K_α line and continuous X-ray spectrum's minimum wavelength ?

(A) remains constant

(B) becomes more than two times

(C) becomes half

(D) becomes less than two times

Q.9

Frequency of radiations arising from lamb shift in a hydrogen atom is 1057 MHz. Identify which part of electromagnetic spectrum does this frequency lie in.

(A) Infrared rays

(B) $\mathrm{X}-$ rays

(C) $\gamma-$ rays

(D) Radio waves

Q.10

When the voltage applied to an X-ray tube increased from $V_1 = 15.5\ \mathrm{kV}$ to $V_2 = 31\ \mathrm{kV}$, the wavelength interval between the K_α line and the cut-off wavelength of the continuous X-ray spectrum increases by a factor of 1 .3. Identify the element.

(Take: $hc = 1240\ \mathrm{eV}\ \ \mathrm{nm}$ and $R = 1 \times 10^7\ \mathrm{N\ m^{-1}}$)

(A) Iron

(B) Manganese

(C) Nickel

(D) Tin

Q.11

Cathode rays of velocity $10^6\ \mathrm{m\ s^{-1}}$ describe an approximately circular path of radius 1 m in an electric field $300\ \mathrm{V\ m^{-1}}$. If the velocity of cathode rays are doubled. The value of electric field so that the rays describe the same

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circular path, will be

(A) $2400\ \text{V m}^{-1}$

(B) $600\ \text{V m}^{-1}$

(C) $1200\ \text{V m}^{-1}$

(D) $12000\ \text{V m}^{-1}$

Q.12

Magnetic field in an electromagnetic wave has maximum value of $20\ \text{nT}$. Then what will be the maximum value of electric field strength?

(A) $9\ \text{V m}^{-1}$

(B) $12\ \text{V m}^{-1}$

(C) $3\ \text{V m}^{-1}$

(D) $6\ \text{V m}^{-1}$

Q.13

A plane electromagnetic wave of wavelength λ has an intensity I. It is propagating along the positive $Y-$direction. The allowed expressions for the electric and magnetic fields are given by:

(A) $\overrightarrow{E} = \sqrt{\frac{2\,I}{\varepsilon_0 c}}\cos\left[\frac{2\pi}{\lambda}(y-ct)\right]\hat{\text{k}},$

$\overrightarrow{B} = +\frac{1}{c}E\hat{\text{i}}$

(B) $\overrightarrow{E} = \sqrt{\frac{I}{\varepsilon_0 c}}\cos\left[\frac{2\pi}{\lambda}(y-ct)\right]\hat{\text{k}},$

$\overrightarrow{B} = +\frac{1}{c}E\hat{\text{i}}$

(C) $\overrightarrow{E} = \sqrt{\frac{2\,I}{\varepsilon_0 c}}\cos\left[\frac{2\pi}{\lambda}(y+ct)\right]\hat{\text{k}},$

$\overrightarrow{B} = +\frac{1}{c}E\hat{\text{i}}$

(D) $\overrightarrow{E} = \sqrt{\frac{I}{\varepsilon_0 c}}\cos\left[\frac{2\pi}{\lambda}(y-ct)\right]\hat{\text{i}},$

$\overrightarrow{B} = +\frac{1}{c}E\hat{\text{k}}$

Q.14

Assertion: Gamma rays are more energetic than X-rays.

Reason: Gamma rays are of nuclear origin but X- rays are produced due to sudden deceleration of high energy electrons while falling on a metal of high atomic number.

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(A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(B) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(C) If Assertion is true but the Reason is false.

(D) If Assertion is false but Reason is true.

Q.15

In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{\mathrm{I}}{2}$, the potential difference V is increased to $2\ \mathrm{V}$, and the separation distance d is reduced to $\frac{\mathrm{d}}{2}$, then

(A) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

Q.16

Which of the statements below given are true ?
A. The wavelength of microwaves is greater than that of $UV - rays$.

B. The wavelength of $IR - rays$ is lesser than that of $UV - rays$.

C. The wavelength of microwaves is lesser than that of $IR - rays$.

D. Gamma rays have shortest wavelength in the Electromagnetic Spectrum.

(A) A

(B) B

(C) C

(D) D

Q.17

Identify the quantity from the given options which is/are transported by an electromagnetic wave

(A) energy

(B) charge

(C) momentum

(D) information

Q.18

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Which among the following represent Maxwells equation?

(A) $\oint \overrightarrow{E}.\overrightarrow{dA} = \frac{q}{\varepsilon_0}$

(B) $\oint \overrightarrow{B}.\overrightarrow{dA} = 0$

(C) $\oint \overrightarrow{E}.\overrightarrow{dl} = \frac{-dB}{dt}$

(D) $\oint \overrightarrow{B}.\overrightarrow{dl} = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

Q.19

Ozone layer above earth's atmosphere will

(A) Prevent infrared radiations from sun reaching on earth

(B) Prevent infrared radiations originated from earth from escaping earth's atmosphere

(C) Prevent ultra violet rays from sun

(D) Reflect back radio waves

Q.20

The electric field in an electromagnetic wave is given by $E = \left(50\ \mathrm{NC}^{-1}\right)\ \sin\omega\left(t - \frac{x}{c}\right)$ Find the intensity of the wave in wm^{-2}.

Answers & Solutions

Q.1 Answer:

$1.4\ \mathrm{kV/m}$

Solution:

$I = \frac{P}{A} = \frac{1}{2}\varepsilon_0 E_0^2 c$

$\therefore E_0 = \sqrt{\frac{2P}{\varepsilon_0 cA}} = \sqrt{\frac{2\times\left(27\times10^{-3}\right)}{\left(9\times10^{-12}\right)\times\left(3\times10^{8}\right)\times\left(10\times10^{-6}\right)}}$

$\Rightarrow E_0 = 1.4\ \mathrm{kV/m}$

Q.2 Answer:

On increasing potential difference between filament and target, frequency of x-rays increases

Solution:

On increasing potential difference between filament and target, frequency of x rays increases, On increasing filament current intensity of xrays increases.

Q.3 Answer:

$\frac{169\ \mathrm{m}}{2q\mathrm{E}_0}$

Solution:

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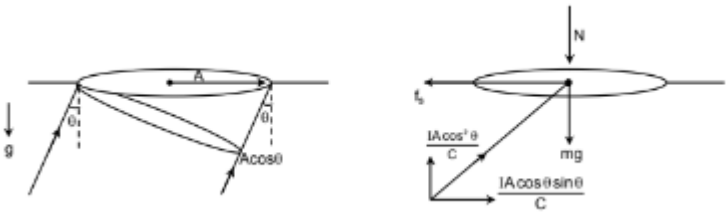
Since magnetic field does not do any work on the charge particle work is being done by only electric field hence

$$qEx = \frac{1}{2}mv^2$$
$$x = \frac{1}{2}\frac{mv^2}{qE} = \frac{169m}{2qE}$$

Q.4 Answer:

8

Solution:



$$\mathrm{N} = \frac{\mathrm{IA}\cos^2\theta}{\mathrm{C}} - \mathrm{mg} \quad \mathrm{f_s} = \frac{\mathrm{IA}\cos\theta\sin\theta}{\mathrm{C}}$$

$$= \frac{\mathrm{IA}}{2\mathrm{C}} - \mathrm{mg} \quad \mathrm{f_s} = \frac{\mathrm{IA}}{2\mathrm{C}}$$

$$\mathrm{f_s} \le \mu \mathrm{N}$$

$$\frac{\mathrm{IA}}{2\mathrm{C}} \le \mu\left(\frac{\mathrm{IA}}{2\mathrm{C}} - \mathrm{mg}\right); \mu \ge \frac{\mathrm{IA}}{\mathrm{IA} - 2\mathrm{mgC}}$$

Q.5 Answer:

0.6 N

Solution:

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$\overrightarrow{B} = B_0\hat{i}[\cos(kz - \omega t)] + B_1\hat{j}\cos(kz + \omega t)$

$B_0 = 3 \times 10^{-5}\,T \quad B_1 = 2 \times 10^{-6}\,T$

Electric filed associated with it is,

$\overrightarrow{E} = -B_0c\hat{j}\cos(kz - \omega t) - B_1c\hat{i}\cos(kz + \omega t)$

Here, c is speed of light in vacuum.

At $z = 0$,

$F_{rms} = \sqrt{\frac{(QB_0c)^2 + (QB_1c)^2}{2}}$

Here c **is speed of light in vacuum.**

$= \sqrt{\frac{(10^{-4} \times 3 \times 10^{-5} \times 3 \times 10^8)^2 + (10^{-4} \times 2 \times 10^{-6} \times 3 \times 10^8)^2}{2}}$

$= \sqrt{\frac{0.81 + 0.0036}{2}} = 0.6\,N$

Q.6 Answer:

$20 \times 10^{-8}\,N$

Solution:

The pressure exerted by the photons

$P = 0.75 \times \frac{I}{c} + 0.25 \times \frac{2I}{c}$ **(** P **= pressure,** c **= speed of light,** I **= intensity)**

$P = 1.25 \times \frac{I}{c}$

We know the force can be computed using

$F = P \times A$ **(** F **= force,** P **= pressure,** A **= area)**

$F = 1.25 \times \left(\frac{50 \times 1}{3 \times 10^8}\right)$

$F = 20.83 \times 10^{-8}\,N$

Q.7 Answer:

15

Solution:

$n\,e\,E = 6\pi\,\eta\,r\,v$ **or** $n = \frac{6\pi\,\eta\,r\,v}{e\,E}$

$= \frac{6 \times 3.14 \times 1.6 \times 10^{-5} \times 5 \times 10^{-7} \times 0.01}{1.6 \times 10^{-19} \times 6.28 \times 10^5} = 15$

Q.8 Answer:

becomes less than two times

Solution:

λ_{min} **is minimum wavelength of continuous** X**-ray**

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$\lambda_{min} = \frac{hc}{E}$

Energy, $E = eV$

$\lambda_{min} = \frac{hc}{eV}$ **.....eq** 1

where, V **is the voltage applied**

$\Delta\lambda = \lambda_{K_\alpha} - \lambda_{\min}$ **.....eq** 2

$\lambda_{min} = \lambda_{K_\alpha} - \Delta\lambda$**.....eq** 3

Now, if V **is halved,** λ_{K_α} **will not be changed as the energy between the two shells remain same and** λ_{min} **becomes doubled from eq** 1

Now, $\Delta\lambda\prime = \lambda_{K_\alpha} - \lambda\prime_{min}$

$\Delta\lambda\prime = \lambda_{K_\alpha} - \lambda\prime_{min}$

$\Delta\lambda\prime = \lambda_{K_\alpha} - 2\lambda_{min}$

From eq 3 **putting in above equation**

$\Delta\lambda\prime = \lambda_{K_\alpha} - 2(\lambda_{K_\alpha} - \Delta\lambda)$

$\Delta\lambda\prime = 2\Delta\lambda - \lambda_{K_\alpha}$

So, $2\Delta\lambda > \Delta\lambda\prime$

Q.9 Answer:

Radio waves

Solution:

This given frequency corresponds to the radio waves i.e short wavelength or high frequency.
Radio waves are a type of electromagnetic radiation with wavelengths longer than infrared light in the electromagnetic spectrum. Charged particles undergoing acceleration, such as time-varying electric currents, produce radio waves. Transmitters create artificial radio waves, which radio receivers receive through antennas. In modern technology, radio waves are commonly used in fixed and mobile radio transmission, television, radar and radio navigation systems, communications satellites, cellular computer networks, and a variety of other applications.

Q.10 Answer:

Iron

Solution:

The minimum wavelength at a given voltage is, $\lambda_{th} = \frac{hc}{eV_a}$
As the applied voltage becomes double the initial, the threshold wavelength becomes half $\frac{\lambda_{\text{th}}}{2}$**.**
From Rydberg's formula to find the wavelength of a spectral line,

$\frac{1}{\lambda_{K\alpha}} = R(z-1)^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$

$\frac{13}{10}(\lambda_{K\alpha} - \lambda_{th}) = \left(\lambda_{K\alpha} - \frac{\lambda_{th}}{2}\right)$

$\frac{3}{10}\lambda_{K\alpha} = \left(\frac{13}{10} - \frac{1}{2}\right)\lambda_{th}$

$\frac{3}{10}\left(\frac{4\times10^{-7}}{3(Z-1)^2}\right) = \left(\frac{8}{10}\right)\frac{12.4\times10^{-7}}{15.5\times10^3}$

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$\Rightarrow \frac{5000}{8} = (Z-1)^2$

$625 = (Z-1)^2 \qquad \Rightarrow Z = 26$

This gives the element Iron.

Q.11 Answer:

1200 V m^{-1}

Solution:

Cathode rays are composed of electrons, when they move in electric field a force

$$F = eE \quad \text{...(i)}$$

Acts on them, this provides the necessary centripetal force to the particles

$F = \frac{mv^2}{r}$ **...(ii)**

From Eqs. (i) and (ii), we get

$eE = \frac{mv^2}{r}$

$$\Rightarrow \qquad r = \frac{mv^2}{eE} = \frac{m(10^6)^2}{e(300)} \quad \text{...(iii)}$$

When velocity is doubled same circular path is followed, hence radius is same

$r = \frac{m(2\times10^6)^2}{eE}$ **...(iv)**

Equating Eqs. (iii) and (iv), we get

$m \times \frac{(10^6)^2}{300e} = \frac{m\times(2\times10^6)^2}{eE}$

$$\Rightarrow \qquad E = 300 \times 4 = 1200 \text{ V cm}^{-1}$$

Q.12 Answer:

6 V m^{-1}

Solution:

Maximum value of electric field $\left|\overrightarrow{E}\right| = \left|\overrightarrow{B}\right| \cdot \left|\overrightarrow{C}\right| = 20 \times 10^{-9} \times 3 \times 10^8$

$= 6 \text{ V/m}$

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Q.13 Answer:

$\overrightarrow{E}=\sqrt{\frac{2I}{\varepsilon_0 c}}\cos\left[\frac{2\pi}{\lambda}(y-ct)\right]\hat{k},$

$\overrightarrow{B}=+\frac{1}{c}E\hat{i}$

Solution:

If E_0 is the amplitude of the electric field then the intensity of the electromagnetic wave is given as $I=\frac{1}{2}\varepsilon_0 E_0^2\times c$. From the above relation, we get $E_0=\sqrt{\frac{2I}{c\,\varepsilon_0}}$. The relation between electric and magnetic fields is $B_0=\frac{E_0}{c}$. The direction of $\overrightarrow{E}\times\overrightarrow{B}$ will be along $+\hat{j}$.

Q.14 Answer:

If both Assertion and Reason are true but Reason is not explanation of the Assertion.

Solution:

If both Assertion and Reason are true but Reason is not explanation of the Assertion.
Because the frequency(v) of gamma rays is greater than X rays, the relation between energy(E) and frequency(v) is, $E=hv$, where h is the Planck's constant.

Q.15 Answer:

the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

Solution:

$$\lambda_{\min}=\frac{hc}{eV}$$
$$\Rightarrow \lambda_{\min}\alpha\frac{1}{V}\Rightarrow(\lambda_{\min})_{new}=\frac{\lambda_2}{2}$$
$$\because I=\frac{dN}{dt}\times\frac{hc}{\lambda}$$
$$\because \frac{dN}{dt}\text{ decreases}$$
$$\text{Hence I decreases}$$

Q.16 Answer:

A

Solution:

The wavelength order of the given types of waves are given below

Waves Wavelength Range (in meter)

Gamma rays $10^{-14}-10^{-10}$

IR-rays $7\times10^{-7}=10^{-3}$

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UV-rays $10^{-9} - 4 \times 10^{-7}$

Microwave $10^{-4} - 10^{0}$

Hence, statements (A) and (D) are correct.

Q.17 Answer:

energy

Solution:

Electromagnetic waves bring energy from one location to another and thus momentum is carried along with it as well.
Furthermore, electromagnetic waves are not deflected by electric field or magnetic field, which indicates that they have no charge whatsoever.
Therefore, EM waves do not carry any charge. The charge is often linked with the mass. Since EM waves are simply waves, they have no mass and therefore no charge.

Q.18 Answer:

$\oint \overrightarrow{E}.\overrightarrow{dA} = \frac{q}{\varepsilon_0}$

Solution:

Maxwell's equations are as follows

(i) $\oint \overrightarrow{E}.\overrightarrow{dA} = \frac{q}{\varepsilon_0}$ **(Gauss's law of electricity)**

(ii) $\oint \overrightarrow{B}.\overrightarrow{dA} = 0$ **(Gauss's law of magnetism)**

(iii) $\oint \overrightarrow{E}.\overrightarrow{dl} = -\frac{d\phi_E}{dt}$ **(Faraday's law)**

(iv) $\oint \overrightarrow{B}.\overrightarrow{dl} = \mu_0 I_C + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$ **(Ampere - Maxwell law)**

Q.19 Answer:

Prevent infrared radiations from sun reaching on earth

Solution:

Ozone layer will absorb UV rays; reflects the infrared radiation and does not reflected back radiowaves

Q.20 Answer:

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Solution:

$I = \frac{1}{2}\varepsilon_0 E_0^2 C$

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